Extending the Relative Approach toward Teaching Effectiveness for Studying and Predicting Educational Equity Gaps

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***Abstract*— Equity gaps refer to the disparities in educational outcomes across race (ethnicity), socioeconomic status, gender, physical/mental abilities, and other demographic traits. In USA, the gaps grew significantly during the COVID-19 pandemic and presented a major challenge in education. In search for a solution to address this challenge proactively and early in the learning process, in this paper, we extend an existing relative approach on the modeling and prediction of teaching effectiveness (*T-Matrices Rev*) into one that can be used to study and predict educational equity gaps (*T-Matrices Rev II)*. The proposed approach represents a practical and applicable methodical tool leveraging machine learning techniques for assisting the research on learning, inferring, and addressing the differences in students' learning outcomes that may have been caused by their different demographic statuses. The proposed research is in its inception stage, and it faces many challenges, which are also discussed in this paper.**

***Keywords*— teaching effectiveness modeling and predication involving equity gaps, applied machine learning, equity gaps in COVID-19**

I. Introduction

In education, equity gaps refer to the disparities in educational outcomes across race/ethnicity, socioeconomic status, gender, physical or mental abilities, and other demographic traits. [23, 24] The term was originally proposed to study the disparities in learning results by demographically diverse students in K-12 schools. Eventually, colleges across America gradually accepted the term while acknowledging the existence of the gaps in college education, which they have been trying to close by investing on the related research on and polices to close the gaps. As a matter of fact, the educational equity gaps in America have kept growing rather than closing. [25] It was further worsened by the COVID-19 pandemic. [26, 27] While many of the previous research on educational equity gaps focused on finding and dealing with the possible causes, there is a lack of a modern data driven approach that leverages the power of artificial intelligence and mathematical modeling. In this paper, we attempt to propose a data driven and machine learning based approach to study and predict educational equity gaps.

A. Background of Educational Assessment and Modeling

In educational assessment [1,2,3,4,5,6,7,8], empirical data are systematically documented and analyzed in order to draw from them the insights on how to help students learn better. There are two types of assessment: direct assessment using data and indirect assessment using inferences from data. Currently, the most common practice for evaluation of teaching effectiveness is course evaluation, which is a direct assessment by students (and often faculties). Its evaluation metrics are composed of grading fairness, lecturing skills, care to students, clarity of presentation, availability for help, lecture preparations, instruction language/modulation/pace, enthusiasm, use of technologies, presentation skills, and student engagement, etc. However, this summative assessment has been constantly and widely criticized for being an inaccurate and unfair measurement on teaching [9, 10,11,12,13,14,15,16].

A mathematics-specific framework for teaching evaluation was proposed, and it was found useful but ineffective at handling not-mathematics-specific concepts [17]. Later on, a synergic collaboration on researching teaching quality was proposed and formed. However, there was still a lack of a common language delineating teaching effectiveness [18]. Reflections on teaching quality also propelled researchers to rethink its definition, and it was found that quality of teaching is also dependent on the characteristics of the student body (audience or teaching targets) [19]. *T-Matrices* [20] has been proposed based on the idea of relative teaching effectiveness [19] for the modeling, assessment, and prediction of teaching effectiveness. However, there are some issues in it [22]. The issues were addressed in [22], which proposed its new version, *T-Matrices Rev*. The effectiveness of *T-Matrices Rev* [22] on teaching effectiveness prediction is illustrated by a case study, which showed promising results.

B. Motivation for Adopting T-Matrices Rev for Studies on Equity Gaps

T-Matrices Rev is a very flexible mathematical framework, into which it is straightforward to build additional perspectives regarding learning effectiveness. Thus, working on *T-Matrices Rev*, we propose *T-Matrices Rev II*, which is an extended version of the original for research on educational equity gaps.

The rest of the paper is organized as follows. Section 2 introduces *T-Matrices Rev II*. Section 3 shows how it can be used to study and predict students' equity gaps. Section 4 discusses the major challenges for this research and concludes this paper.

II. The Theoretical Framework

In *T-Matrices Rev II*, student and course are considered a collection of elements. The elements of a course represent the set of its learning objectives.

$$C = (o^1, o^2, \dots, o^n), o^i \text{ is a learning objective of } C.$$

A student is represented as two sets of properties. The first set, Set P, represents a student's properties unrelated to demographic traits. The second set, Set D, represents his/her properties related to demographic traits

$$S = (p^1, p^2, \dots, p^l, d^1, d^2, \dots, d^k),$$

$$p^i \text{ is a property of } S, \text{and } d^i \text{ is a demographic property of } S.$$

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For example, one demographic property d^m of a student could be defined as having one of the following values, 1, 2, and 3, for depicting his/her gender of "Male", "Female", and "Other Gender". Another demographic property d^n of a student could be defined as having one of the different values depicting his/her racial category.

All or part of Set P properties may be referenced by a course for determining the student's academic performance. Since we assume that demographic traits may also play a role in determining the performance, Set D shall also be referenced by a course for the purpose of determining learning outcomes.

$$S_C = (p^1, p^2, \dots, p^m, d^1, d^2, \dots, d^k),$$

$$p^i \text{ is a property of } S \text{ that is relevant to } C \text{ and}$$

$$\text{all of } S\text{'s demographic properties.}$$

We represent them as vectors. In addition, the CO vector stands for the weights of the learning outcomes for a course.

$$S_c{}^T = \begin{bmatrix} p^1 \\ . \\ p^m \\ d^1 \\ . \\ d^k \end{bmatrix} \quad C^T = \begin{bmatrix} o^1 \\ o^2 \\ . \\ . \\ . \\ o^n \end{bmatrix} \quad CO^T = \begin{bmatrix} co^1 \\ co^2 \\ . \\ . \\ . \\ co^n \end{bmatrix}, \ where \sum_i co^i = 1$$

A teacher's effectiveness of teaching is defined as his/her ability to transform a student's course specific properties S_c into the related academic performance in terms of completion rates of the course's learning outcomes. We represent the effective as the following.

$$E_{TC}{}^{\alpha}, where\ \alpha \text{ denotes } T\text{'s teaching mode}$$

In addition, we also acknowledge that an seasoned teacher may have different ways to teach a course. So we assume it is possible that a teacher shall have different related teaching modes for a course. We use a superscript alpha to demote a specific teaching mode of a teacher for a course. The exact definition of the effective is given as follows.

$$E_{TC}{}^{\alpha} = \begin{bmatrix} E_{TC}^{11} & E_{TC}^{21} & . & E_{TC}^{n1} \\ E_{TC}^{12} & E_{TC}^{22} & . & E_{TC}^{n2} \\ . & . & . & . \\ . & . & . & . \\ . & . & . & . \\ E_{TC}^{1m} & E_{TC}^{2m} & . & E_{TC}^{nm} \\ E'^{11}_{TC} & E'^{21}_{TC} & . & E'^{n1}_{TC} \\ E'^{12}_{TC} & E'^{22}_{TC} & . & E'^{n2}_{TC} \\ . & . & . & . \\ . & . & . & . \\ . & . & . & . \\ E'^{1k}_{TC} & E'^{2k}_{TC} & . & E'^{nk}_{TC} \end{bmatrix}, \ where \begin{cases} n = |C| \\ m + k = |S_C| \end{cases}$$

In the above definition, the value of n stands for the number of learning objectives in a curse C and that of m+k stands for the number of properties in S_C. Each component (cell in the matrix) of $E_{TC}{}^{\alpha}$ stands for a function that takes a property of S_C as an input. A teacher teaching a course C to a student S is defined as a process that translate the various properties of S_c into the various degrees of completeness of (and thus the teaching results on) all of the course's learning objectives.

$$S_c \circ E_{TC}{}^{\alpha} = \begin{bmatrix} E_{TC}^{11}(p^1) + & . & + E_{TC}^{1m}(p^m) + E'^{11}_{TC}(d^1) + & . & E'^{1k}_{TC}(d^k) \\ E_{TC}^{21}(p^1) + & . & + E_{TC}^{2m}(p^m) + E'^{21}_{TC}(d^1) + & . & E'^{2k}_{TC}(d^k) \\ . & . & . & . & . & . \\ . & . & . & . & . & . \\ . & . & . & . & . & . \\ E_{TC}^{n1}(p^1) + & . & + E_{TC}^{nm}(p^m) + E'^{n1}_{TC}(d^1) + & . & E'^{nk}_{TC}(d^k) \end{bmatrix}^T$$

$$= R^S_{T_\alpha C} = \begin{bmatrix} q^1 \\ q^2 \\ . \\ . \\ . \\ q^n \end{bmatrix}^T, \ where\ q^i \in [0,1]$$

The quality of teaching (learning outcomes) is defined as the normalized teaching results.

$$Q^C_{T_\alpha}(S_c) = R^S_{T_\alpha C} \cdot CO^T$$

For a group of students **S**, the collective teaching result is defined as follows. Such a collective result is given in a vector (or Matrix) form.

$$\boldsymbol{S_c} = \begin{bmatrix} S_c{}^1 \\ S_c{}^2 \\ . \\ . \\ . \\ S_c{}^k \end{bmatrix} \quad \boldsymbol{R^S_{T_\alpha C}} = \boldsymbol{S_c} \circ E_{TC}{}^{\alpha} \cdot CO^T = \begin{bmatrix} Q^C_T(S_c{}^1) \\ Q^C_T(S_c{}^2) \\ . \\ . \\ . \\ Q^C_T(S_c{}^k) \end{bmatrix}$$

Subsequently, the teaching effectiveness for a class of students is defined as the average of the results.

$$Q^C_{T_\alpha}(\boldsymbol{S_c}) = \frac{\sum_{S_c} Q^C_{T_\alpha}(S_c)}{|\boldsymbol{S_c}|}, \quad where\ S_c \in \boldsymbol{S_c}$$

III. Prediction on Equity Gaps

A. Training a Teaching Simulation Engine with Machine Learning

For a teacher T teaching a course C using a specific teaching mode alpha, assume that we have collected enough data on the teaching results (from the corresponding teaching history documents) for a set of students **S**. Leveraging machine learning techniques, we can train a simulation engine (software) that simulates that teacher's teaching of the course as follows.

$$Sim_E_{TC}{}^{\alpha} = ML_Training(T, \alpha, C, \boldsymbol{S_c}, \boldsymbol{R^S_{T_\alpha C}})$$

B. Prediction of Teaching Results

Using the teaching simulation software, we can predict the teaching results (quality) of the corresponding teacher on the course using a specific mode (alpha) for a specific student.

$$Sim_Q^C_{T_\alpha}(S_c) = R^S_{T_\alpha C} \cdot CO^T = S_c \circ Sim_E_{TC}{}^{\alpha} \cdot CO^T$$

C. Using Different Samples to Predict Equity Gaps

A benefit of the teaching results prediction is that we can use different sample data (simulating students with different sets of demographic traits) to obtain their predicted teaching results (qualities) and thus compare the results to study the equity gaps. Suppose we have two students *Student_A* and *Student_B*. They are from different demographic backgrounds, and hence they have different values for their Set D properties. Then we can use the simulated teaching results (qualities) to obtain the difference to study the related equity gap problems.

$$Sim_Q^C_{T_\alpha}(Student_A_c)\ vs.\ Sim_Q^C_{T_\alpha}(Student_B_c)$$

IV. Disscusions and Conclusions

The above relative approach on teaching effectiveness to study equity gap problems represents an effort to tackle the equity gap research using a more practical approach leveraging machine learning. It is a research direction that is

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still in the inception stage. In reality, there are many challenges that must be met to provide sufficient conditions for this research to continue to develop and evolve.

A. Data Collection and Privacy Protection

Obviously, the biggest challenge for this research is on data collection, as the major data that we need to collect for this research are considered private data about students (both the demographic and academic data). In US, the Family Educational Rights and Privacy Act (FERPA) forbids the collection of sensitive student data including the demographic data and those from students' academic records. However, it provides two exceptions. Exception 1 states that with the consent of a student, his/her data could be used by a third party. Exception 2 states that for academic purpose, student academic record data could be used, provided that certain precautions are taken as well as with an institutional consent as well.

However, as a matter fact, due to the sensitiveness of the issue on collecting student data, the vast majority of American institutions tend to be reluctant to endorse it for the purpose of research like ours, worrying about possible negative consequences. Talking to students and hence obtaining their consents to using their data seems to be the only viable choice, but it is extremely inefficient when considering the vast amount of data that we need to collect for our research. In addition, more often the credibility of the data collected from voluntary student participants are also not without problems.

Similar laws and regulations on student data collection also exist in Europe, China, and the other major regions.

To meet this challenge, we strongly urge the governmental agencies and academic institutional societies to make and enact new laws and/or regulations to legalize/endorse appropriate usages of students' data.

B. What Demographic Properties of Students to Collect?

While different sources may use/share the same set of definitions on demographic properties for studies, there has not been a technically complete agreement on what composes the demographic information. In addition, the granularities of some of the possible demographic property also present challenges. For example, is "Chinese" a proper granularity for the racial property of a Chinese student, while there could be more than 50 minorities (sub-races) that fall in this major racial category?

Similarly, to meet this challenge, we propose to limit the research to the regions where there is an established consensus on the range and granularity of demographic properties. Only when the research in this direction become mature enough in different regions, a common standard on them could established.

C. Representation of The Nature of Teaching

The major technical issue of the research shall be on the representation of teaching E_{TC}^{α}. Is it true that the teaching results of a learning outcome is determined by the linear composition of all the results by the functions mapping each student property to a value? The answer is that it may not necessary be true, and more research work shall be done to determine the internal mechanism (and hence the corresponding mathematical form) on how the student properties affect their learnings on a course taught by a certain teacher.

Luckily, the are many scientometric, machine learning, and statistical methods that we can use to produce the teaching simulation engine without specifying the exact ways of the mappings. The representation given for E_{TC}^{α} in this paper stands for an initial attempt and it will not impact the training of the teaching simulation engine (using supervised learning). We think that it could be discovered later (with more data and experiments) that the representation should be more like a networked structure.

D. Implementation of T-Matrices Rev II and Future Work

The implementation of *T-Matrices Rev II* as a software package is currently underway. Specifically, we are in the phase of collecting domain requirements. We feel cautiously confident that the first running version of this software will be available in the latter half of the next year.

In the future, we would like to explore adding into *T-Matrices Rev II* the culture and domain specific modules to enrich, illustrate, and better define the related demographic properties of students.

In all, the proposed research represents a practical semi-formal methodical tool levering machine learning to study and predict educational equity gaps. It is highly applicable. In the future, we would like to explore this research by looking for alternatives that will help us meet the above-mentioned major challenges.

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